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Advanced Drug Delivery Reviews 36 (1999) 229–254

advanced
drug delivery
reviews

Recent advances in the brain targeting of neuropharmaceuticals by chemical delivery systems

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Abstract

Brain-targeted chemical delivery systems represent a general and systematic method that can provide localized and sustained release for a variety of therapeutic agents including neuropeptides. By using a sequential metabolism approach, they exploit the specific trafficking properties of the blood–brain barrier and provide site-specific or site-enhanced delivery. After a brief description of the design principles, the present article reviews a number of specific delivery examples (zidovudine, ganciclovir, lomustine benzylpenicillin, estradiol, enkephalin, TRH, kyotorphin), together with representative synthetic routes, physicochemical properties, metabolic pathways, and pharmacological data. A reevaluated correlation for more than 60 drugs between previously published in vivo cerebrovascular permeability data and octanol/water partition coefficients is also included since it may be useful in characterizing the properties of the blood–brain barrier, including active transport by P-glycoprotein. © 1999 Elsevier Science B.V. All rights reserved.

Keywords: Blood–brain barrier; Brain-targeted redox analog; Cyclodextrin; Estradiol; Neuropeptides; Octanol–water partition; P-glycoprotein; Retrometabolic design; Zidovudine (AZT)

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1. Introduction

The brain is a delicate organ, and evolution built very efficient ways to protect it. Unfortunately, the same mechanisms that protect it against intrusive chemicals can also frustrate therapeutic interventions. Many existing pharmaceuticals are rendered ineffective in the treatment of cerebral diseases by our inability to effectively deliver and sustain them within the brain. General methods that can enhance drug delivery to the brain are, therefore, of great interest. The present article intends to review recent advances in the field of brain-targeted chemical delivery systems, a rational drug design approach that exploits sequential metabolism not only to deliver, but to target, drugs at their site of action. By localizing drugs at their desired site of action one can reduce toxicity and increase treatment efficiency. To illustrate the complexity of the problems that have to be overcome for successful brain targeting, a brief characterization of the blood–brain barrier (BBB) is also included.

2. The blood–brain barrier

2.1. Structural and enzymatic aspects

It is now well established that the blood–brain barrier is a unique membranous barrier that tightly segregates the brain from the circulating blood [1–8]. Capillaries of the vertebrate brain and spinal cord lack the small pores that allow rapid movement of solutes from circulation into other organs; these capillaries are lined with a layer of special endothelial cells that lack fenestrations and are sealed with tight junctions. Tight epithelium, similar in nature to this barrier, can also be found in other organs (skin, bladder, colon, lung) [9]. These endothelial cells, together with perivascular elements such as astrocytes and pericytes, constitute the blood–brain

barrier. Neuronal endings may also directly innervate the endothelium [10]. It seems that a barrier is present only in animals with a complex level of higher neural function, and qualitative observations suggest that ‘higher’ functions of neural tissue are protected by a tighter barrier [11]. Nevertheless, certain areas within the brain (circumventricular organs) are ‘outside the barrier’ in the sense that they lack a tight epithelium [11].

Microvessels make up an estimated 95% of the total surface area of the BBB, and they represent the principal route by which chemicals enter the brain [12]. Vessels in brain were found to have somewhat smaller diameters and thinner walls than vessels in other organs [13]. Also, the mitochondrial density in brain microvessels was found higher than in other capillaries not because of more numerous or larger mitochondria, but because of the small dimensions of the brain microvessels and the consequently smaller cytoplasmic area [13]. In brain capillaries, intercellular cleft, pinocytosis, and fenestrae are virtually nonexistent; exchange must pass transcellularly. Therefore, only lipid-soluble solutes that can freely diffuse through the capillary endothelial membrane may passively cross the BBB. In general capillaries, such exchange is overshadowed by other nonspecific exchanges. Despite the estimated total length of 650 km and total surface area of 12 m² of capillaries in human brain [10], this barrier is very efficient and makes the brain practically inaccessible for lipid-insoluble compounds, such as polar molecules and small ions. As a consequence, the therapeutic value of many promising drugs is diminished, and cerebral diseases proved to be most refractory to therapeutic interventions. Given the prevalence of brain diseases alone, this is a considerable problem. A 1988 study found the 1-month prevalence of mental disorders for over 18-year-olds in the US to be 15.4%: 5.1% affective disorders, 7.3% anxiety disorders, 3.8% substance abuse/dependence, and 1.3% severe cognitive impairment [14]. Practically all drugs

currently used for disorders of the brain are lipid-soluble and can readily cross the BBB following oral administration.

As so many things in modern medicinal chemistry, the concept of a barrier between blood and brain can be traced back to the work of Ehrlich, who noticed in 1885 that in laboratory animals i.v. injection of dyes such as trypan blue colored all tissues except the brain [15]. Regarding the exact nature and localization of this barrier, a long period of uncertainty followed. During this period, in 1933 Spatz was the first to suggest the capillary endothelium as an essential structure of this barrier [16], and later Crone also argued in favor of a peculiar tightness of the cerebral endothelium [17]. Nevertheless, only Reese and Karnovsky's 1967 experiment with horse-radish peroxidase [18] provided a convincing proof favoring this idea.

The blood–brain barrier also has an additional, enzymatic aspect: solutes crossing the cell membrane are subsequently exposed to degrading enzymes present in large numbers inside the endothelial cells that contain large densities of mitochondria, metabolically highly active organelles. Enzymes and receptors found in the BBB include, among others, adenylate cyclase, guanylate cyclase, $\text{Na}^+ - \text{K}^+$ ATPase, alkaline phosphatase, catechol O-methyl transferase (COMT), monoamine oxidase (MAO), GABA transaminase, DOPA decarboxylase [5,9,19]. Blood–brain barrier enzymes also recognize and rapidly degrade most naturally occurring neuropeptides [20].

There are a number of other aspects, too. In 1946, Krogh, while pointing to lipid solubility as an important factor determining access to the brain, already raised the possibility of some sort of active transport processes in the endothelium of the vessels in the central nervous system (CNS) [21]. It is now well known that a few special solutes have access to specific, catalyzed transport; for example, glucose transporter ensures that glucose, an essential nutrient, reaches the brain despite its low lipid solubility [22,23]. Carriers are facilitative in nature and have been identified for hexoses, neutral, acidic, and basic amino acids, monocarboxylic acids, nucleosides and nucleoside bases, choline, and other biologically important molecules. These protein macromolecular systems are characterized by saturability and molecu-

lar selectivity. This avenue provides for the movement of selected molecules, but, with some exceptions, these systems are not relevant to drug delivery.

More recently, active efflux pumps have also been identified. The existence of probenecid-sensitive, active pumps for organic anions, such as beta lactam drugs or valproic acid, has been suspected for a while [24], and evidence for their existence is accumulating [25,26]. As an even more important development, it also has become evident that P-glycoprotein, a multidrug transporter, is present in the BBB. It has relatively low substrate specificity and actively excludes a number of lipophilic compounds from cerebral endothelial cells [8,27–31]. P-glycoprotein is a membrane protein belonging to the ABC (ATP-binding cassette) transport protein family, but contrary to most other members, it has apparently wide substrate specificity. It seems that substrates, or at least some of them, are pumped out from the membrane itself [32]. Fig. 1 gives a schematic representation of all these BBB properties using a comparison between brain and general capillaries.

2.2. Octanol–water partition coefficients

The log octanol–water partition coefficient ($\log P_{o/w}$) probably still represents the most informative physicochemical parameter used in medicinal chemistry and countless examples where it proved a useful descriptor are available in the literature [33–37]. Selection of the octanol/water system as reference is usually rationalized as modeling partition between aqueous and biophases [33,38]. Octanol, with a polar head and a flexible, nonpolar tail, has hydrogen-bonding capabilities and amphiphilicity characteristics similar to those of the phospholipids and proteins found in biological membranes.

In a work that underscores again the suitability of the octanol/water system to characterize CNS effects, Franks and Lieb [39] found that, for anesthetic potency plotted against solvent/water partition coefficient, octanol gave an excellent correlation for all compounds and all animals over an extremely wide range. Correlations were much better than for oil/water, hexadecane/water, or benzene/water systems—even if, for example, benzene has a solubility parameter (δ) not very different from octanol (9.2

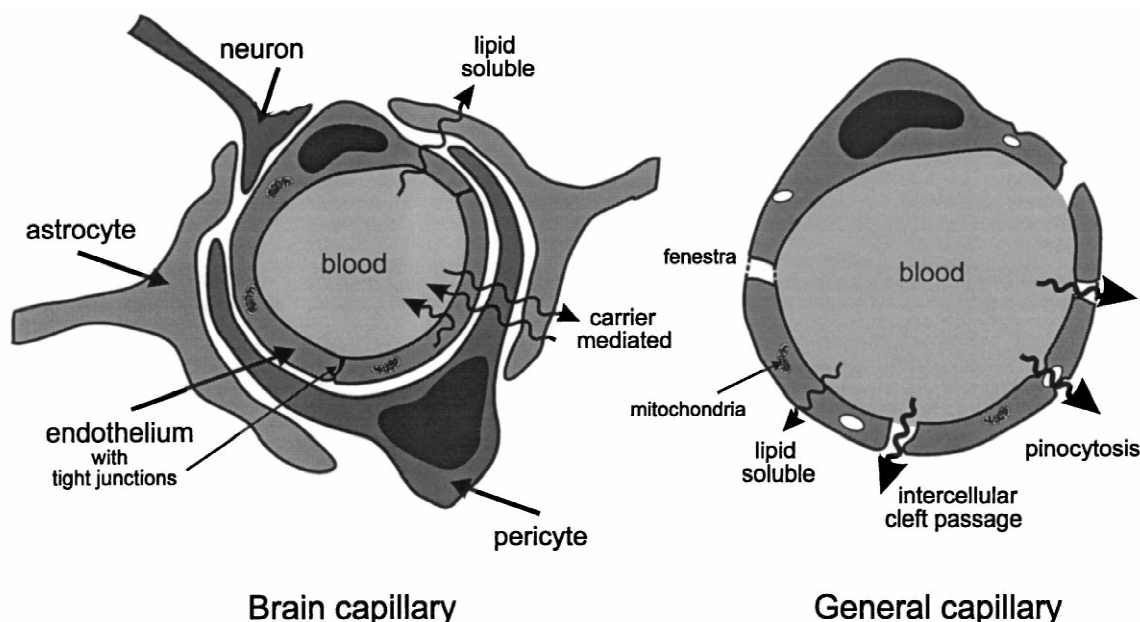


Fig. 1. Schematic comparison between general and brain capillaries. See text for details.

vs. $10.3 \text{ cal}^{1/2} \text{ cm}^{-3/2}$) [40]. As general anesthetics were found to have no effect on lipid bilayer structures when studied using X-ray and neutron diffraction, the authors even concluded that this suggests that a single site of action is sufficient to account for this data, and this site has solubility characteristics similar to those of octanol.

However, $\log P_{o/w}$ seems to have a limited performance in predicting brain/blood concentration ratios ($\log \text{BB}$) [41–44]. The octanol/water system still has no serious rival, but alternative partition systems that might give a better description have been suggested and investigated. Notable attempts include the use of cyclohexane/water [45]; the definition of a ‘critical quartet’ of octanol (amphiprotic), alkane (inert), chloroform (proton donor), and propylene glycol dimer (proton acceptor) [46]; or the use of $\Delta \log P$ between octanol/water and cyclohexane/water [41,47,48]. More recently, it has been suggested that with correction terms $\log P_{o/w}$ may still provide a reasonable description of $\log \text{BB}$ [44]. Nevertheless, this parameter may prove useful even in describing active transport through the BBB. Contrary to $\log \text{BB}$, in vivo $\log$ cerebrovascular permeability data tend to correlate much better

with $\log P_{o/w}$. Some lipophilic compounds, however, showed an unexpectedly low apparent transfer rate. This has been interpreted in terms of high binding to plasma proteins and/or of a threshold of molecular weight cut-off for transfer through BBB [49,50]. However, as suggested recently by Tsuji and Tamai [30], active efflux by P-glycoprotein may also account for this behavior.

For a more accurate interpretation, we present here a comprehensive and reevaluated correlation between $\log P_{o/w}$ and in vivo permeability data of rat brain capillaries estimated from the decrease of arterial plasma concentration (corrected for plasma protein binding). As Fig. 2 indicates, there is a good correlation over a wide range of about 8 log units. The figure was obtained based on $\log P_{o/w}$ data from the original publications or, whenever possible, recommended values from Hansch et al. [37] and permeability data collected from several sources [49–54]. If needed for conversion, a specific surface area of $100 \text{ cm}^2/\text{g}$ was used for the rat BBB in vivo [4,50]. For compounds where more permeability coefficient (P_c) values were available, the average value was used (e.g., antipyrine, mannitol, sucrose, thiourea, urea). The permeability coefficient of pro-

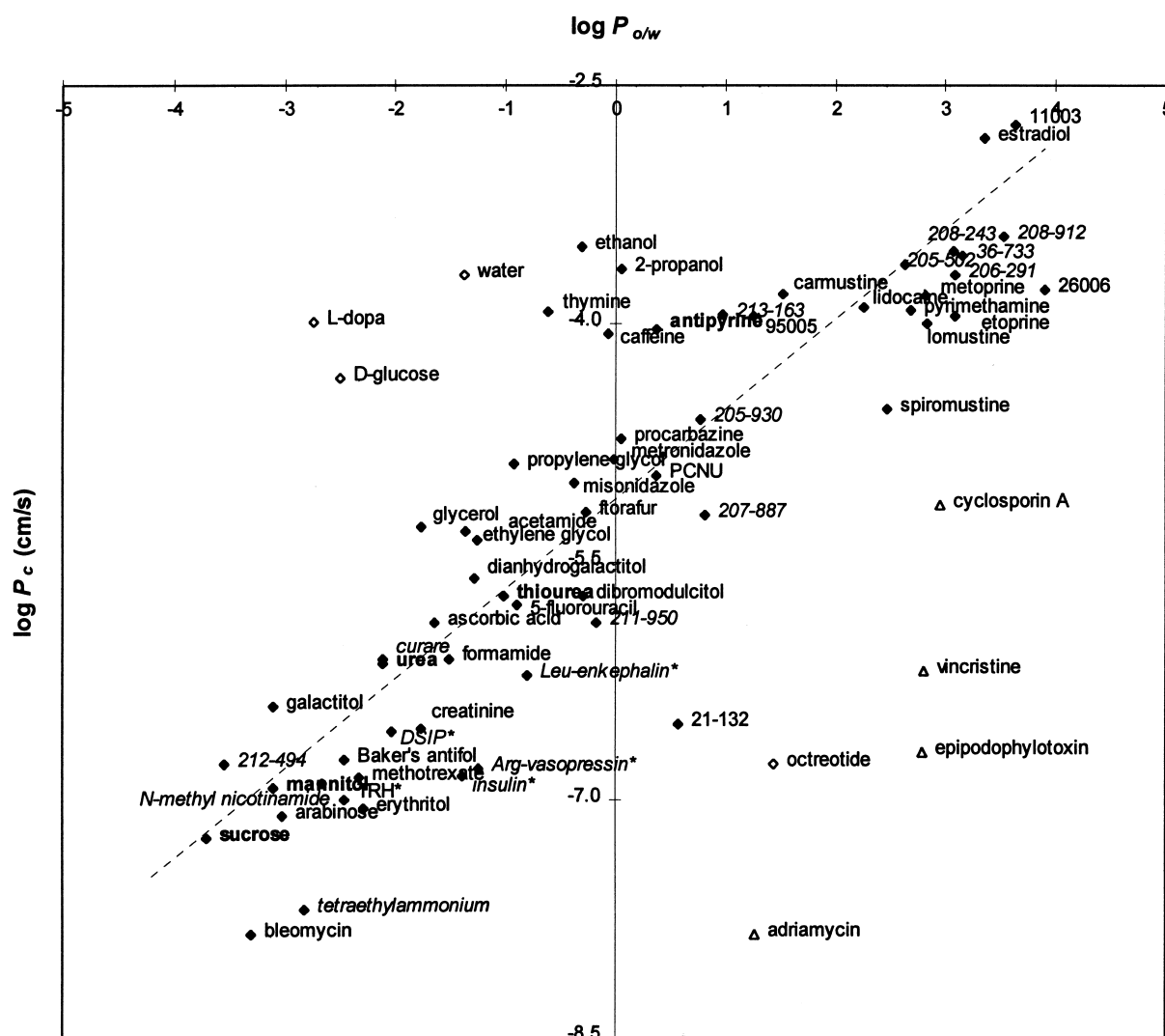


Fig. 2. Correlation between the in vivo log permeability coefficient of rat brain capillaries ($\log P_c$) and log octanol/water partition coefficient ($\log P_{o/w}$). For compounds in *italics*, the log distribution coefficient measured at physiological pH was used ($\log D_{o/w}$). Values denoted with * are for guinea pig. Practically all strong deviants below the line (denoted with triangles) have been shown recently to be substrates for P-glycoprotein, a multidrug transporter that actively removes them from the brain.

pranolol [54], somewhat higher than predicted by this trend, was omitted. For a few molecules, many of whom do not have a neutral form, we included apparent partition coefficients measured around physiological pH if they were available (indicated in *italics* in Fig. 2). Most previous works included a molecular weight correction ($MW^{-1/2}$), related to the diffusion coefficient. Here we omitted this term,

since, with the present data, $\log P_{o/w}$ alone gives similarly good linearity. Some compounds, such as cyclosporin A, vincristine, adriamycin, or epipodophylotoxin, indeed fell below the trendline, but it is now well established that practically all of them are substrates of P-glycoprotein, a multidrug transporter that actively excludes them from cerebral endothelial cells [8,27–31]. If these deviations are

indeed mostly due to P-glycoprotein activity, then octreotide, which also fell below the line, is a likely P-glycoprotein substrate as well.

Omitting the obvious outliers, for the remaining 58 compounds one obtains:

$$\log P_c = -5.102(\pm 0.081) + 0.566(\pm 0.038)\log P_{o/w}$$

$$n = 58, \quad r = 0.896, \quad \sigma = 0.617, \quad F = 226.8 \quad (1)$$

This is a reasonably good correlation, and it is derived from considerably more data than those previously published. As a comparison, Rapoport [51] had $n=17$, Levin [49] had $n=22$ and $r=0.91$, Hansch [55] had $n=23$ and $r=0.93$, Pardridge [50] had $n=13$ and $r=0.84$, and Gratton [54] had $n=18$ and $r=0.88$.

2.3. Drug delivery to the brain

It is clear from the previous sections that delivery of poorly lipid-soluble compounds to the brain requires some way of getting past the blood–brain barrier. There are several possible strategies, such as transient osmotic opening of the BBB, exploiting natural chemical transporters, high-dose chemotherapy, or even biodegradable implants. But all of these methods have major limitations: they are invasive procedures, have toxic side effects and low efficiency, and are not sufficiently safe.

Another possible strategy is to smuggle compounds across as their lipophilic precursors. Heroin, a diacyl derivative of morphine, is a notorious example that crosses the BBB about 100 times more easily than its parent drug just by being more lipophilic. Similarly, brain uptake of drugs can be improved via prodrug formation [56–58]. Prodrugs are pharmacologically inactive compounds that result from transient chemical modifications of biologically active species. The chemical change is usually designed to improve some deficient physicochemical property, such as membrane permeability or water solubility. After administration, the prodrug, by virtue of its improved characteristics, is brought closer to the receptor site and is maintained there for longer periods of time. Here it can convert to the active form, usually via a single activating step. For example, esterification or amidation of hydroxy-, amino-, or carboxylic acid-containing drugs, may

greatly enhance lipid solubility and, hence, entry into the brain. Once in the CNS, hydrolysis of the modifying group will release the active compound.

Unfortunately, simple prodrugs suffer from several important limitations. While increased lipophilicity may improve movement across the BBB, it also tends to increase uptake into other tissues causing an increased tissue burden. This nonselectivity in delivery is especially detrimental when potent drugs such as steroids or cytotoxic agents are considered, since toxicity is exacerbated at nontarget sites. Furthermore, while increased lipophilicity may facilitate drug uptake into the CNS, it also enhances efflux processes. This can result in poor tissue retention and short biological action. Finally, while the only metabolism associated with a prodrug should be its conversion to the parent drug, other routes can occur, and the formed metabolites may contribute to the toxicity of the compounds. These effects, that is poor selectivity, poor retention, and the possibility for reactive metabolites, may often conspire to decrease, not to increase, the therapeutic index of drugs masked as prodrugs.

3. Chemical delivery systems

It was partly to address this challenge that our research group began developing novel chemical delivery systems (CDSs) starting in the early 1980s. The cunning aspect of these brain-targeted systems is that, in addition to providing access by increasing the lipophilicity, they exploit the specific properties of the BBB to ‘lock’ drugs in the brain on arrival and prevent them from recrossing the BBB. Today, CDS design has evolved into a broad subject and, together with soft drug design [59,60], is integrated within so-called retrometabolic drug design approaches. Retrometabolic drug design approaches are systematic methodologies aimed to improve the therapeutic index that thoroughly integrate structure–activity and structure–metabolism relationships (SAR/SMR) in the drug design process [61–63].

Chemical delivery systems represent novel and systematic ways of targeting active biological molecules to specific target sites or organs based on predictable enzymatic activation. They are inactive chemical derivatives of a drug obtained by one or

more chemical modifications so that the newly attached moieties are monomolecular units (generally comparable in size to the original molecule) and provide a site-specific or site-enhanced delivery of the drug through multistep enzymatic and/or chemical transformations. Some of these delivery systems have been reviewed previously [5,63,64].

During the chemical manipulations, two types of bioremovable moieties are introduced to convert the drug into an inactive precursor form. A targetor (T) moiety is responsible for targeting, site-specificity, and lock-in, while modifier functions ($F_1 \dots F_n$) serve as lipophilizers, protect certain functions, or fine-tune the necessary molecular properties to prevent premature, unwanted metabolic conversions. The CDS is designed to undergo sequential metabolic conversions, disengaging the modifier functions and finally the targetor, after this moiety fulfils its site- or organ-targeting role. Undoubtedly, the concept evolved from the prodrug concept, but became essentially different by the introduction of multistep activation and targetor moieties. Within the present formalism, one can say that prodrugs contain one or more F moieties for protected or enhanced overall delivery, but they do not contain T moieties.

Brain-targeting chemical delivery systems represent just one class of CDSs; however, this is the most developed class. Using the general CDS concept, successful deliveries have been achieved to the brain, to the eye, and to the lung as summarized recently [63]. Here, we will include a more detailed description of brain-targeting CDSs, and we will review some recent developments within this field.

4. Brain-targeting chemical delivery systems

4.1. Design principles

These CDSs are based on the idea that, if a lipophilic compound that enters the brain is converted there into a lipid-insoluble molecule, it will no longer be able to come out, it will become 'locked-in'. If the same conversion also takes place in the rest of the body, it accelerates peripheral elimination and improves targeting.

In principle, many targetor moieties are possible for a general system of this kind, but the one based

on the 1,4-dihydrotrigonelline $\leftrightarrow$ trigonelline (cof-fearine) system, where the lipophilic 1,4-dihydro form (T) is converted in vivo to the hydrophilic quaternary form (T^+), proved the most useful. This conversion takes place easily everywhere in the body since it is closely related to that of the ubiquitous NAD(P)H $\leftrightarrow$ NAD(P) $^+$ coenzyme system associated with numerous oxidoreductases and cellular respiration. Since oxidation takes place with direct hydride transfer and without generating highly active or reactive radical intermediates, it provides a nontoxic targetor system. Furthermore, since for small quaternary pyridinium ions rapid elimination from the brain, probably due to involvement of an active transport mechanism that eliminates small organic ions, has been shown [65,66], the T^+ moiety formed during the final release of the active drug D from the charged T^+-D form will not accumulate within the brain.

Meanwhile, the charged T^+-D form is locked behind the BBB into the brain, but is easily eliminated from the body due to the acquired positive charge, which enhances water solubility. After a relatively short time, the delivered drug D (as the inactive, locked-in T^+-D) is present essentially only in the brain, providing sustained and brain-specific release of the active drug. It has to be emphasized that the system not only achieves delivery to the brain, but it provides preferential delivery, which means brain targeting. Ultimately, this should allow smaller doses and reduce peripheral side effects. Furthermore, since the 'lock-in' mechanism works against the concentration gradient, it provides more prolonged effects. Consequently, CDSs can be used not only to deliver compounds that otherwise have no access to the brain, but also to retain lipophilic compounds within the brain, as has indeed been achieved, for example, with a variety of steroid hormones.

During the last decade, the system has been explored with a wide variety of drug classes. The following is certainly not an exhaustive list, but it represents a quite comprehensive and representative collection for future reference—biogenic amines: phenylethylamine [67,68]; steroid hormones: testosterone [69], progestins [70], dexamethasone [71,72], progesterone [73], estradiol [74–79]; anti-infective agents: penicillins [80–83]; antivirals: acyclovir

[64,84], trifluorothymidine [85,86], ribavirin [87–89], Ara-A [87], deoxyuridines [90–92], 2'-F-5-methylarabinosyluracil [93]; antiretrovirals: zidovudine (AZT) [94–106], 2',3'-dideoxythymidine [66], ganciclovir [107], cytosinyl-oxathiolane [108]; anticancer agents: lomustine (CCNU) [109,110], HECNU [110], chlorambucil [111]; neurotransmitters: dopamine [112–115], GABA [116,117]; nerve growth factor (NGF) inducers: catechol derivatives [118,119]; anticonvulsants: GABA [117], phenytoin [120], stiripentol [121], felodipine [122]; monoamine oxidase (MAO) inhibitors: tranylcypromine [123]; anti-dementia drugs (cholinesterase inhibitors): tacrine [124]; antioxidants: LY231617 [125]; nonsteroidal anti-inflammatory drugs (NSAIDs): indomethacin [126], naproxen [126]; anesthetics: propofol [127]; nucleosides: adenosine [128]; and peptides: tryptophan [129], Leu-enkephalin analogue [130,131], thyrotropin-releasing hormone (TRH) analogue [132], kyotorphin analogues [133,134].

In a recent addition to the drug-targeting arsenal, targeted drug delivery to the brain via phosphonate derivatives was also explored, and so-called anionic chemical delivery systems (aCDS) were designed, synthesized, and evaluated for testosterone and AZT [135,136]. Here, an (acyloxy)alkyl phosphonate-type targetor moiety is used, and formation of an anionic intermediate (T^- -D) is expected to provide the 'lock-in'. The following sections provide a number of specific examples for brain-targeting CDSs, together with representative synthetic routes, physicochemical properties, metabolic pathways, and pharmacological data.

4.2. Zidovudine (AZT)-CDS

Infection by human immunodeficiency virus type 1 (HIV-1), the causative agent of AIDS, has many devastating consequences, including an encephalopathy with subsequent dementia. A vast majority of AIDS patients has demonstrable neuro-histological damage related to this syndrome. To adequately treat AIDS dementia, antiviral agents must gain access to the CNS in therapeutically relevant concentrations, but many potentially useful drugs are unable to penetrate into brain tissue. Zidovudine (azidothymidine, AZT) was the first drug approved for the treatment of AIDS. This modified

riboside has been shown to be useful in improving the neuropsychiatric course of AIDS encephalopathy in a few patients, but the doses required for this improvement usually precipitated severe anemia. As this antiviral agent has a primary alcohol functionality in the 5'-position, attachment of the targetor through an ester bond susceptible for hydrolysis can be relatively easily carried out. Together with the increasing occurrence of AIDS, this probably represented the main reason why a number of laboratories started to investigate the AZT-CDS [94–106]. Fig. 3 summarizes a possible synthetic route [95], and is included here to illustrate one of the simplest possible synthetic routes for a CDS. AZT was treated with nicotinoyl chloride hydrochloride in pyridine to yield the 5'-nicotinate. This ester was subsequently quaternized with methyl iodide to yield the 5'-trigonellinate (1-methylnicotinate) (T^+ -AZT) and reduced in basic aqueous sodium dithionite to give the 5'-(1,4-dihydrotrigonellinate) derivative of AZT (AZT-CDS). The obtained AZT-CDS is a crystalline solid, which is stable at room temperature for several months when protected from light and moisture. The CDS is relatively stable in pH 7.4 phosphate buffer, but rapidly oxidizes in enzymatic media. In addition, T^+ -AZT, the depot form of AZT, was shown to gradually release the parent compound.

The ability of AZT-CDS to provide increased brain AZT levels has been demonstrated in a number of different in vivo animal models. In mice, the areas under the curve (AUCs) obtained by Chu and coworkers [105] for AZT concentrations after i.v.

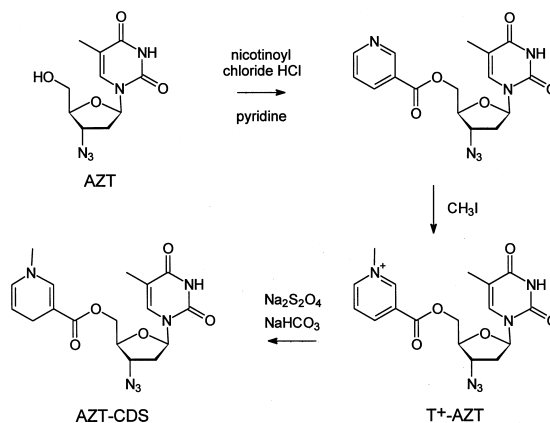


Fig. 3. Synthesis of AZT-CDS. See text for details.

administration of 187 $\mu\text{mol/kg}$ of AZT-CDS/AZT were 11.28/1.21 $\mu\text{g}\times\text{h/ml}$ and 26.64/25.38 $\mu\text{g}\times\text{h/ml}$ in brain and serum, respectively. This means a 9.32 times increase in the relative brain exposure to AZT. Similar data obtained more recently by Miz-rachi and coworkers [97] in rats after i.v. administration of 130 $\mu\text{mol/kg}$ of AZT-CDS/AZT were 7.09/0.23 $\mu\text{g}\times\text{h/g}$ and 9.67/26.33 $\mu\text{g}\times\text{h/g}$ in brain and blood, respectively. This means a 31.51 times increase in the relative brain exposure to AZT and a simultaneous 2.72 times decrease in blood exposure. In rabbits, Brewster and coworkers [96] demonstrated a 3- to 4-fold increase in delivery to the brain. While the brain/blood ratio never exceeded 1 after AZT administration, indicating that the drug is always in higher concentration in the blood than in the brain, it approached 3 at 1 h post-dosing with AZT-CDS. Similarly, in dogs, Brewster and coworkers obtained AZT levels that were 46% lower in blood, but 175–330% higher in brain after AZT-CDS administration compared to parent drug administration [98]. Here, AUCs obtained for AZT concentrations in blood after i.v. administration of 64 $\mu\text{mol/kg}$ of AZT-CDS/AZT were 52.95/98.45 $\mu\text{g}\times\text{h/ml}$. Furthermore, *in vitro* experiments found AZT-CDS not only more effective in inhibiting human immunodeficiency virus (HIV) replication than AZT itself, presumably due to hydrolysis, but also less toxic to host lymphocytes [97,100,101]. In conclusion, a number of different studies found an improved delivery to the brain and a decreased potential for dose-limiting side effects. This suggests that AZT-CDS may become a useful agent in the treatment of AIDS encephalopathy and the related dementia.

4.3. Ganciclovir-CDS

Enhanced brain delivery with a CDS approach has also been achieved recently for ganciclovir, another antiviral agent that may be useful in treating human encephalitic cytomegalovirus (CMV) infection [107]. CMV infections are common and usually benign; however, human CMV occurs in 94% of all patients suffering with AIDS, and it has been implicated as a deadly cofactor. While positive results were achieved in the treatment of the associated retinitis, the treatment of encephalitic cytomegalic disease is far

less successful with ganciclovir. A brain-targeted CDS (Fig. 4) was, therefore, investigated, and improved organ selectivity was demonstrated. In rats, the AUCs obtained by Brewster and coworkers [107] for ganciclovir concentrations after i.v. administration of 80 $\mu\text{mol/kg}$ of ganciclovir-CDS/ganciclovir were 3.61/0.66 $\mu\text{g}\times\text{h/ml}$ and 1.42/10.46 $\mu\text{g}\times\text{h/ml}$ in brain and blood, respectively. This means a 5.44 times increase in the relative brain exposure to ganciclovir and a simultaneous 7.38 times decrease in blood exposure. Fig. 4 illustrates the obtained ganciclovir concentrations in plasma and in brain as a function of time. As can be seen, the CDS can sustain drug levels in the brain at therapeutically significant levels, and it can reduce blood concentrations which are associated with toxicity.

4.4. Lomustine (CCNU)-CDS

The (2-chloroethyl)nitrosoureas are among the most effective antitumor agents and have demonstrated clinical activity against a variety of malignant diseases, in particular brain tumors and lymphomas. Most of these agents are lipid soluble and can cross the BBB, but a variety of physicochemical and metabolic factors causing poor CNS retention and peripheral toxicity complicate their use in the treatment of central cancers. Consequently, a CDS approach that can offer a 'lock-in' mechanism was investigated [109,110].

For the depot form, $\text{T}^+ \text{-CCNU(OH)}$, the AUC ratio between brain and blood measured in rats after i.v. administration of a 200 $\mu\text{mol/kg}$ dose was found to be 50.05, proving a successful 'lock-in' [109]. However, CCNU(OH) levels could not be sustained for sufficiently long periods, probably due to metabolism and relatively high lipophilicity. We included this example here to illustrate two aspects of the general CDS design. One of them is flexibility. Lomustine (CCNU) has no functional group that would allow easy attachment of a targetor function. However, it has an active hydroxy metabolite, *N*-(2-chloroethyl)-*N'*-(*trans*-4-hydroxycyclohexyl)-*N*-nitrosourea (CCNU(OH)), that can be easily used in a CDS design. The second aspect illustrated here is the physicochemical aspect. Since adequate changes in partition properties are crucial for the success of this delivery system, this is a very important aspect, and

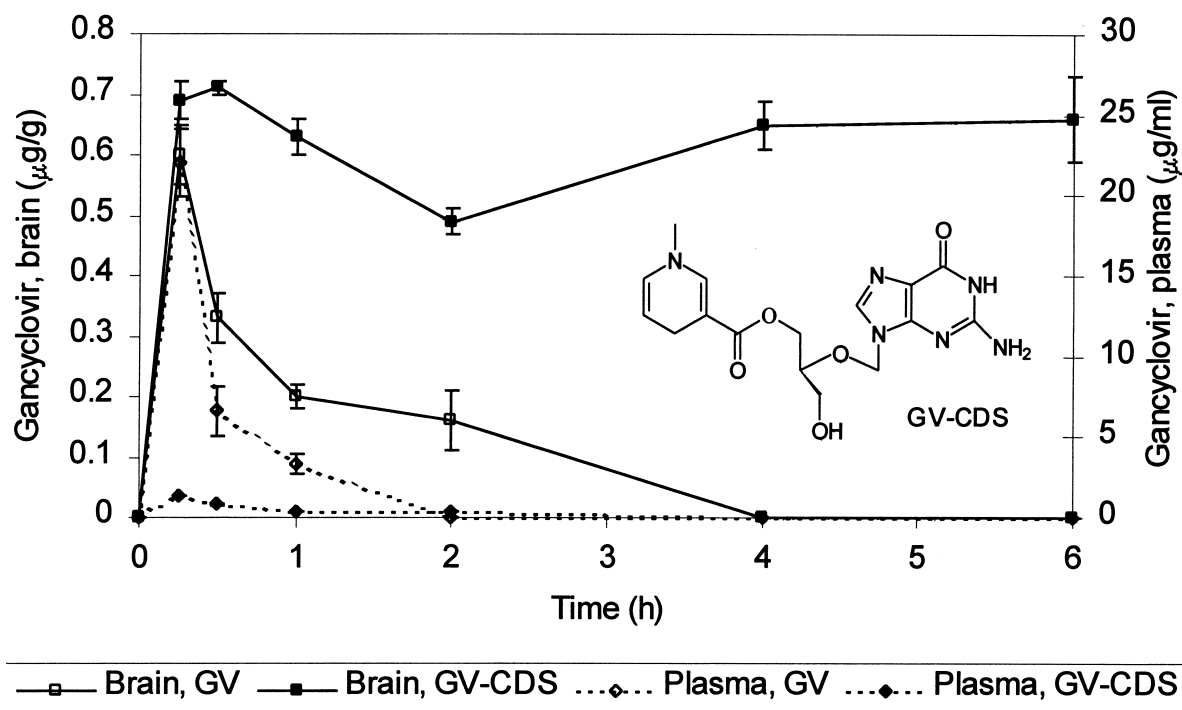


Fig. 4. Ganciclovir (GV) concentrations in brain and plasma as a function of time after an i.v. dose of 80 $\mu\text{mol/kg}$ of GV or GV-CDS. Data are mean $\pm$ SEM in rats.

this class of chemical delivery system is indeed designated as enzymatic physical–chemical-based CDS. Fig. 5 illustrates the ‘lock-in’ mechanism for this system and also summarizes measured and calculated partition coefficients. The parent drug, CCNU(OH), is already lipophilic enough to passively diffuse across the BBB, as indicated by its octanol/water partition coefficient: $\log P_{o/w} = 1.11$, 1.00 [37], 0.99 [109]. Attachment of the 1,4-dihydrotrigonelline targetor further increases lipophilicity, $\log P_{o/w} = 2.53$ [37,109], and should enhance transport across the blood–brain barrier. Oxidation of the targetor moiety leads to an ionic form with a measured octanol/water apparent partition (distribution) coefficient that is about four orders of magnitude smaller: $\log D_{o/w} = -1.30$ [37,109]. Such a significant change in partition properties could easily explain the success of the ‘lock-in’ mechanism. The figure also contains calculated values obtained using a recently developed, molecular size-based method [137–139]. This distinctively simple model can be used to estimate partition properties for

a wide variety of organic molecules including non-zwitterionic peptides. In addition, even if distribution coefficients of charged species are very difficult to predict as they are counterion-dependent, the partition data of quaternary pyridinium-type compounds measured around physiological pH can be reasonably well described [140]. As illustrated in Fig. 5, the method slightly underestimates the CCNU(OH) partition data, but predicts very well the occurring changes.

4.5. Benzylpenicillin-CDS

The treatment of various bacterial infections localized in brain and cerebrospinal fluid is often difficult because β -lactam antibiotics, including penicillins and cephalosporins, poorly penetrate into the CNS. Moreover, since these antibiotics can return from the CNS back to the blood (active transport may be involved), simple prodrug approaches could not provide real solutions. Consequently, a number of CDSs have been synthesized and investigated [80–

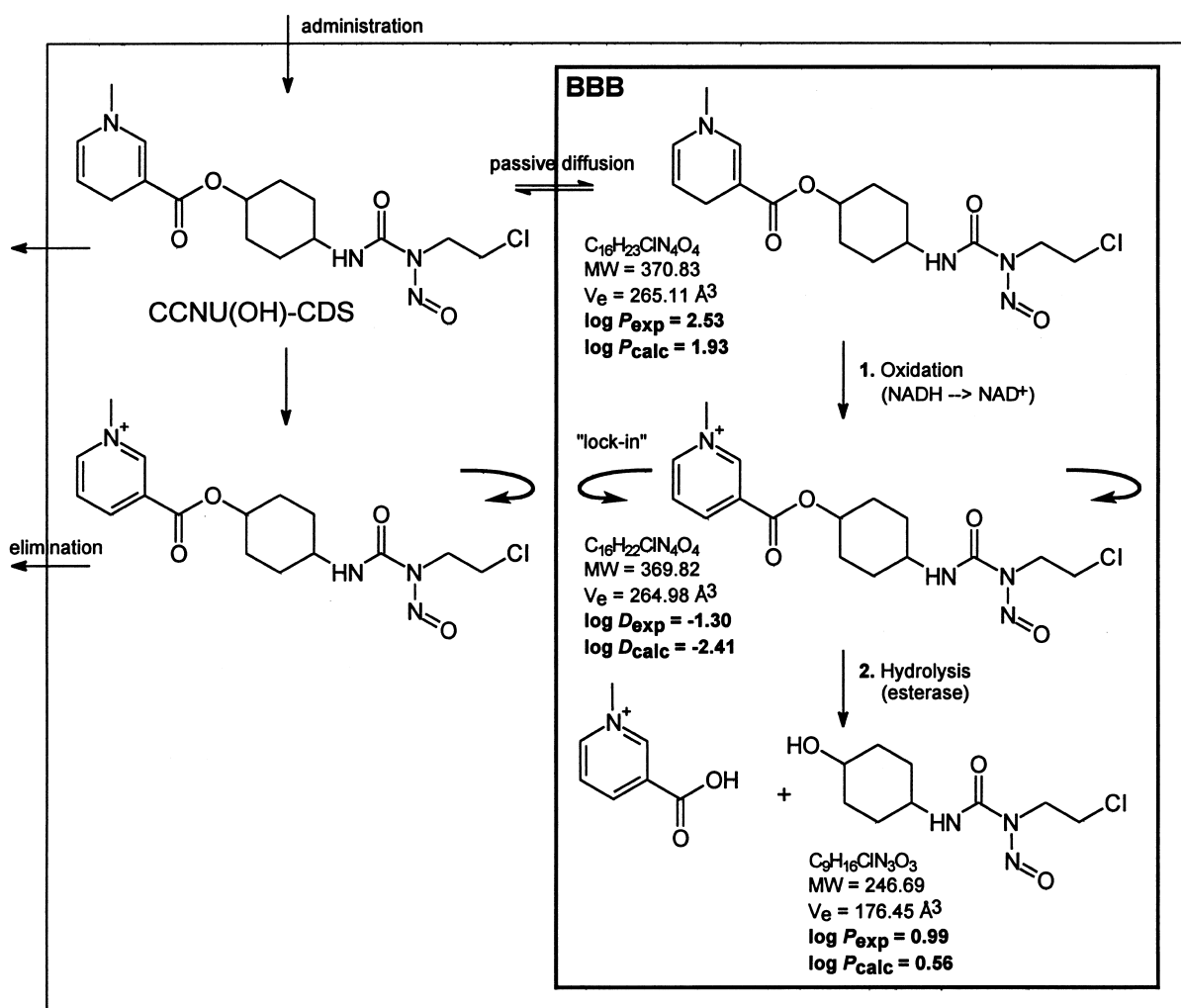


Fig. 5. Illustration of the 'lock-in' mechanism for CCNU(OH)-CDS. Partition coefficients are shown to illustrate the significant changes that occur during the sequential metabolism. While the relatively lipophilic CDS ($\log P > 2$) can easily cross the BBB, the hydrophilic intermediate ($\log D < -1$) is no longer able to come out and can provide a sustained release of the active drug.

83]. Since in this case the coupling of the targetor moiety had to be performed at an acid function, a spacer function had to be inserted between the targetor and the drug. This renders additional flexibility to the approach, and several possibilities were investigated. This was one of the first cases where the influence of such functions on the stability and delivery properties of CDSs could be investigated. Systematic studies of the effects of various modifications have been performed since then [131,141].

In rats, the AUCs obtained by Wu and coworkers

[82] for benzylpenicillin concentrations after i.v. administration of 60 $\mu\text{mol/kg}$ of benzylpenicillin-CDS/benzylpenicillin were 2.23/0.17 $\mu\text{g} \times \text{h/ml}$ in brain. This means a 12.85 times increase in the relative brain exposure to benzylpenicillin. From the several coupling possibilities investigated, diesters of methylene diol and ethylene 1,2-diol proved worthy of further investigation, but ester–amide combinations did not prove successful (AUC 0.03 and 0.01 $\mu\text{g} \times \text{h/ml}$). Consequently, the diesters have also been investigated in rabbits and dogs [83]. Such

studies are needed, because esterases in rodents are more active than esterases of more evolved species, and hydrolysis by these enzymes is an important step in these delivery processes. The stability of acylox-yalkyl-type esters, present in the CDSs, usually increases in the order rat < rabbit < dog < human.

Both in rabbits and in dogs, i.v. administration of equimolar doses of CDSs provided benzylpenicillin levels in brain and cerebrospinal fluid (CSF) that were substantially higher and more prolonged than in the case of parent drug administration. In this study, the diester of methylene diol proved better; the diester of ethylene 1,2-diol released lower drug concentrations due to formation of an inactive hydroxyethyl ester. In rabbits, CDS administration increased benzylpenicillin brain concentrations 3 to 4 times at the 10 and 30 min sampling periods. Because dogs were not sacrificed, only blood and CSF data were collected. According to these, better CSF penetration was achieved in dogs than in rabbits. AUCs obtained for benzylpenicillin concentrations after i.v. administration of 30 $\mu\text{mol/kg}$ of benzylpenicillin-CDS/benzylpenicillin were 5.07/0.16 $\mu\text{g}\times\text{h/ml}$ and 11.75/24.52 $\mu\text{g}\times\text{h/ml}$ in CSF and blood, respectively. This means a 31.87 times increase in the relative CSF exposure to benzylpenicillin and a simultaneous 2.08 times decrease in blood exposure.

It should be also noted that while the parenteral injection of benzylpenicillins may sometimes cause undesired side effects, such as seizures, the high CNS levels obtained in dogs and rabbits after CDS administration were not accompanied by any toxic side effects [83]. Similarly, while the delivered compound exhibited toxicity in the rotorod test at 0.5 h following a 300 mg/kg i.p. dose (lethal to 4/4 mice), felodipine-CDS displayed no toxicity at either 0.5 h (0/8) or 4 h (0/4) after a 100 mg/kg i.p. dose or 0.5 h (0/4) for a 300 mg/kg i.p. dose [122]. Some toxicity (1/2), which is not surprising considering the toxicity of the delivered compound itself, was observed at 4 h for the 300 mg/kg CDS dose.

4.6. Estradiol-CDS

Among all CDSs, estradiol-CDS is in the most advanced investigation stage, and it is currently undergoing phase I and II clinical trials. Estrogens

are lipophilic steroids that are not impeded in their entry to the CNS. They can readily penetrate the BBB and achieve high central levels after peripheral administration. Unfortunately, estrogens are poorly retained within the brain. Therefore, in order to maintain therapeutically significant concentrations, frequent doses have to be administered. Constant peripheral exposure to estrogens has been related, however, to a number of pathological conditions including cancer, hypertension, and altered metabolism [142,143]. As the CNS is the target site for many estrogenic actions, brain-targeted delivery may provide safer and more effective agents. Estrogen CDSs could be useful in reducing the secretion of luteinizing hormone-releasing hormone (LHRH) and, hence, in reducing the secretion of luteinizing hormone (LH) and gonadal steroids. As such, they could be employed to achieve contraception and to reduce the growth of peripheral steroid-dependent tumors, such as those of the breast, uterus, and prostate, and to treat endometriosis. They also could be useful in stimulating male and female sexual behavior, and in the treatment of menopausal vasomotor symptoms ('hot flashes') [144]. Other potential uses are in neuroprotection, in the reduction of body weight, or in the treatment of depression and various types of dementia, including Alzheimer's disease [145].

Estradiol (E_2) is the most potent natural steroid. It contains two hydroxy functions: one in the phenolic 3 position and one in the 17 position. With these synthetic handles, three possible CDSs can be designed attaching the targetor at the 17-, at the 3-, or at both positions. Attachment at either position, but especially at the 17 position, should greatly decrease the pharmacological activity of E_2 , as these esters are known not to interact with estrogen receptors [146].

Since the first synthesis of E_2 -CDS in 1986 [74], it has been investigated in several models [75–79,147,148]. In vitro studies using rat organ homogenates as the test matrix indicated half-lives of 156.6 min, 29.9 min, and 29.2 min for E_2 -CDS (T at the 17 position) in plasma, liver, and brain homogenates, respectively [74]. Thus, E_2 -CDS is converted to the corresponding quaternary form (T^+-E_2) faster in the tissue homogenates than in plasma. This is consistent with the hypothesis of a membrane-bound enzyme,

such as the members of the NADH transhydrogenase family, acting as oxidative catalyst. These studies also indicated a very slow production of E_2 from T^+-E_2 , suggesting a possible slow and sustained release of estradiol from brain deposits of T^+-E_2 . To detect doses of E_2 -CDS, T^+-E_2 , and E_2 of physiological significance, a selective and sensitive method was needed. This problem was solved using a precolumn-enriching HPLC system [76] that allowed accurate detection in plasma samples and organ homogenates with limits of 10, 20, and 50 ng/ml or g for T^+-E_2 , E_2 -CDS, and E_2 , respectively. This study proved that, in rats, E_2 released from the T^+-E_2 intermediate formed after i.v. E_2 -CDS administration has an elimination half-life of more than 200 h (Fig. 6) and brain E_2 levels are elevated 4 to 5 times longer after E_2 -CDS administration than after simple estradiol treatment [76]. Studies in orchidectomized rats proved that a single i.v. injection of E_2 -CDS suppressed LH secretion by 88, 86, and 66% relative to DMSO controls at 12, 18, and 24 days, respectively, and that E_2 levels were not elevated relative to the DMSO control at any sampling time [148]. A large number of other very encouraging results have been obtained in various animal models and phase I/II clinical trials; most of them have been reviewed previously [64,77]. Fig. 7 presents the effect of i.v. and buccal E_2 -CDS relative to oral Prodynova® (estradiol valerate, 1.0 mg) on LH suppression and on E_2 serum elevation in a study of 18 post-menopausal volunteers. Clinical evaluations suggest a potent central effect with only marginal elevations in systemic estrogen levels; therefore, E_2 -CDS may become a useful and safe therapy for menopausal symptoms or for estrogen-

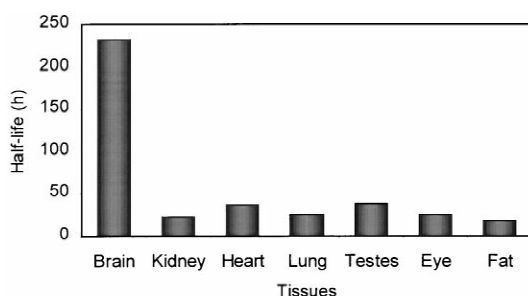


Fig. 6. Elimination half-lives in various tissues for the T^+-E_2 formed after i.v. administration of 38.1 μ mol/kg E_2 -CDS in rats.

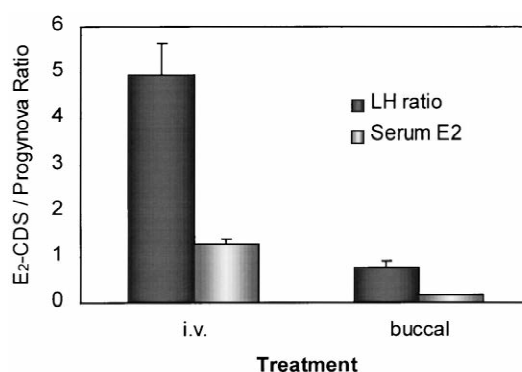


Fig. 7. The effect of i.v. or buccal E_2 -CDS relative to oral Prodynova® (estradiol valerate) on LH suppression or E_2 serum elevation in a study of 18 post-menopausal volunteers. Data are collected from all dosing groups and are corrected for baseline and dose.

dependent cognitive effects. Recently, E_2 -CDS also was shown to provide very encouraging neuroprotective effects. In ovariectomized rats, pretreatment with E_2 -CDS decreased the mortality caused by middle cerebral artery (MCA) occlusion from 65 to 16% [78]. Even when administered 40 or 90 min after MCA occlusion, E_2 -CDS reduced the area of ischemia by 45 to 90% or 31%, respectively. Another recent study provided evidence that treatment with E_2 -CDS can protect cholinergic neurons in the medial septum from lesion-induced degeneration [79].

4.7. Cyclodextrin complexes

Unfortunately, the same physicochemical characteristics that allow for successful chemical delivery also complicate the development of acceptable pharmaceutical formulations. The increased lipophilicity allows partition into deep brain compartments, but also confers poor aqueous solubility. The oxidative lability, which is needed for the 'lock-in' mechanism, and the hydrolytic instability, which releases the modifier functions or the active drug, combine to limit the shelf-life of the CDS. Cyclodextrins may provide a possible solution. They are torus-shaped oligosaccharides that contain various numbers of α -1,4-linked glucose units (6, 7, and 8 for α -, β -, and γ -cyclodextrin, respectively). The number of units determines the size of a cone-like cavity (Fig. 8) into

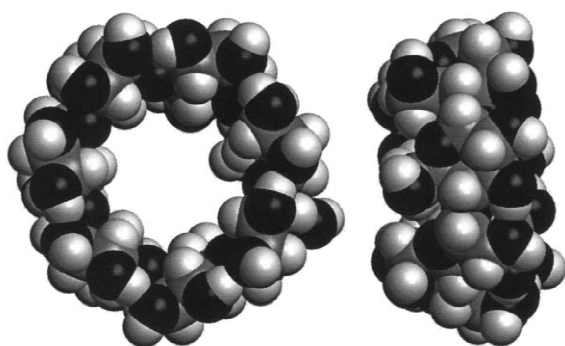


Fig. 8. Two perpendicular views of a CPK structure showing the cavity inside the torus-shaped structure of β -cyclodextrin as determined in a crystal of $\beta\text{CO} \cdot 11\text{H}_2\text{O}$ by neutron diffraction. Atomic coordinates used are those given by Betzel and coworkers (J. Am. Chem. Soc. 106 (1984) 7545–7557). All secondary hydroxyl groups are along the wider edge, which is on the front and right side, respectively, and all primary hydroxyl groups are along the narrower edge. Oxygens are shown in black, carbons in gray, and hydrogens in light gray.

which various compounds can include and form stable complexes [149]. During the last decade, the usefulness of cyclodextrin inclusion complexes in improving the pharmaceutical characteristics of various drugs has become well established [150]. There are a number of potential advantages such as enhancement of solubility, stabilization of hydrolytically or otherwise unstable drugs, enhancement of bioavailability, transformation of liquids into powders, improvement of smell or taste characteristics by prevention of evaporation, reduction of possible stomach injuries, inhibition of hemolysis, and many others.

The corresponding inclusion complex with 2-hydroxypropyl- β -cyclodextrin (HP β CD) solved essentially all problems with E_2 -CDS [151]. This modified cyclodextrin was selected based on its low toxicity observed using various administration routes and based on the fact that alkylation or hydroxy-alkylation of the glucose oligomer can disrupt hydrogen bonding and provide increased water solubility for the compound and for its inclusion complexes as well [152]. Indeed, the aqueous solubility of E_2 -CDS was enhanced about 250,000-fold in a 40% (w/v) HP β CD solution (from 65.8 ng/ml to 16.36 mg/ml). The phase solubility diagram indicated that a 1:1 complex forms at low HP β CD concentration but a

1:2 complex occurs at higher HP β CD concentrations. The stability of E_2 -CDS was also significantly increased allowing formulation in acceptable form. The rate of ferricyanide-mediated oxidation, a good indicator of oxidative stability, was decreased about 10-fold, and shelf-life was increased about four-fold, as indicated by t_{90} and t_{50} values in the temperature range 23–80°C [151]. The cyclodextrin complex even provided better distribution by preventing retention of the solid material precipitated in the lung.

Similarly promising results were obtained for lomustine-CDS [110] and for benzylpenicillin-CDS [153]. For the latter, aqueous solubility was enhanced about 70,000-fold in a 20% (w/v) HP β CD solution (from 50–70 ng/ml to 4.2 mg/ml), and stability was also increased: the stability constant (K_c) of the CDS-HP β CD complex was in the range 720–940 M^{-1} .

5. Molecular packaging

5.1. Design principles

The CDS approach has been recently extended to achieve successful brain deliveries of enkephalin, TRH, and kyotorphin analogues [130–134]. Neuropeptides, peptides that act on the brain or spinal cord, represent the largest class of transmitter substance and have considerable therapeutic potential [133,154,155]. The number of identified endogenous peptides is growing exponentially; recently, almost 600 sequences for neuropeptides and related peptides have been listed [156]. However, delivery of peptides through the BBB is an even more complex problem than delivery of other drugs because they can be rapidly inactivated by ubiquitous peptidases. Therefore, for a successful delivery, three issues have to be solved simultaneously: enhance passive transport by increasing the lipophilicity, assure enzymatic stability to prevent premature degradation, and exploit the ‘lock-in’ mechanism to provide targeting. The solution is a complex molecular packaging strategy, where the peptide unit is part of a bulky molecule, dominated by lipophilic modifying groups that direct BBB penetration and prevent recognition by peptidases [157]. Such a brain-

targeted packaged peptide delivery system contains the following major components: the redox targetor (T); a spacer function (S), consisting of strategically used amino acids to ensure timely removal of the charged targetor from the peptide; the peptide itself (P); and a bulky lipophilic moiety (L) attached through an ester bond or sometimes through a C-terminal adjuster (A) at the carboxyl terminal to enhance lipid solubility and to disguise the peptide nature of the molecule (Fig. 9).

To achieve delivery and sustained activity with such complex systems, it is very important that the designated enzymatic reactions take place in a specific sequence. Upon delivery, the first step must be the conversion of the targetor to allow for 'lock-in'. This must be followed by removal of the L function to form a direct precursor of the peptide that is still attached to the charged targetor. Subsequent

cleavage of the targetor–spacer moiety finally leads to the active peptide (Fig. 9).

5.2. *Leu-enkephalin analogue*

Since the discovery of endogenous peptides with opiate activity [158], many potential roles have been identified for these substances. Possible therapeutic applications could extend from treatment of opiate dependence to numerous other CNS-mediated dysfunctions. Opioid peptides are implicated in mediating neuroendocrine activity, behavioral stress responses, eating disorders, alcoholism, schizophrenia, and in modulating memory function, heart rate, blood pressure, gastrointestinal function, sexual behavior and development. However, analgesia remains their best known role, and it is commonly used to evaluate endogenous opioid peptide activity. While

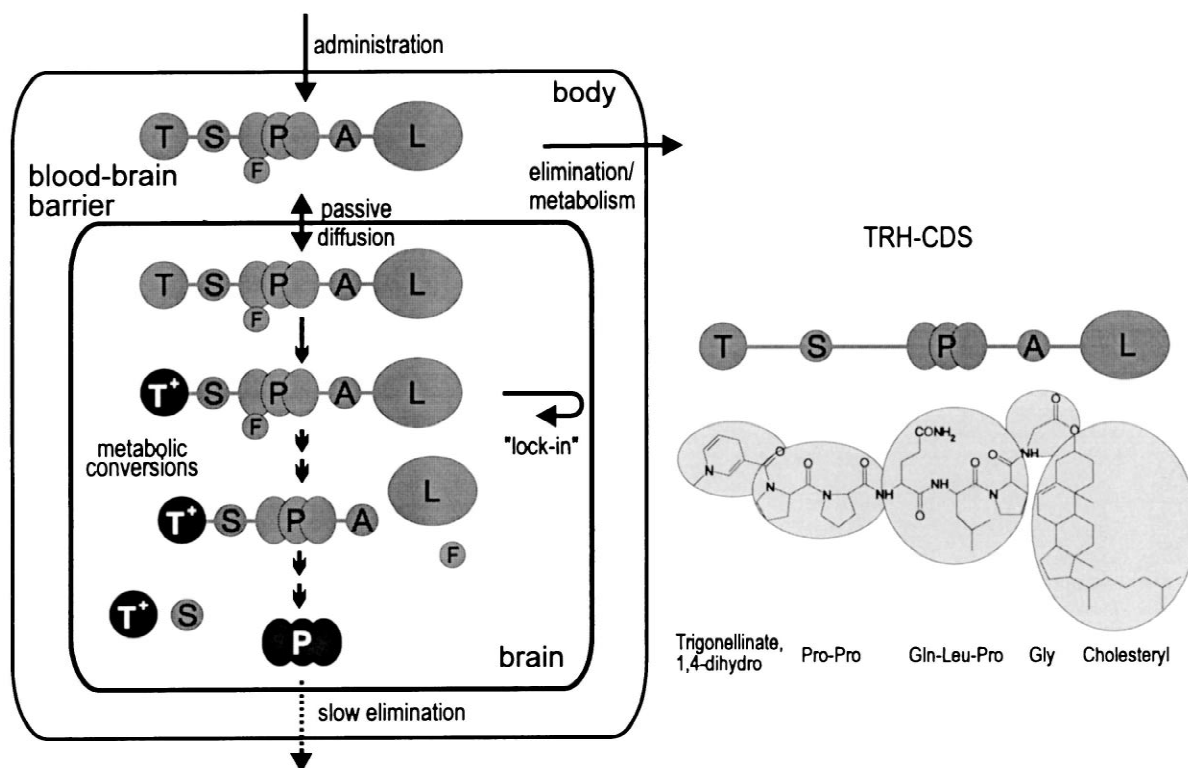


Fig. 9. Schematic representation of the molecular packaging and sequential metabolism used for brain targeting of neuropeptides. TRH-CDS is included to provide a concrete illustration for the targetor (T), spacer (S), peptide (P), adjuster (A), and lipophilic (L) moieties.

native opioid peptides could alter pain threshold after intracerebroventricular (i.c.v.) dosing, they were ineffective after systemic injection. It was reasonable, therefore, to attempt a brain-targeted CDS approach.

In a work considered as one of the top 10 medical advances for 1992 by the *Harvard Health Letter* [159], the first successful delivery and sustained release was achieved for Tyr-D-Ala-Gly-Phe-D-Leu (DADLE), an analogue of leucine-enkephalin, a naturally occurring linear pentapeptide (Tyr-Gly-Gly-Phe-Leu) that binds to opioid receptors [130]. In rat brain tissue samples collected 15 min after i.v. CDS administration, electrospray ionization mass spectrometry clearly showed the presence of the locked-in T^+-D form at an estimated concentration of 600 pmol/g of tissue. The same ion was absent from the sample collected from the control animals treated with the vehicle solution only. To optimize this delivery strategy, an effective synthetic route

was established for peptide CDSs, and the role of the spacer and the lipophilic functions was investigated [131]. Four different CDSs were synthesized by a segment-coupling method. Their i.v. injection produced a significant and long-lasting (>5 h) response in rats monitored by a tail-flick latency test. Compared with the delivered peptide itself, the packaged peptide and the intermediates formed during the sequential metabolism had weak affinity for opioid receptors. The antinociceptive effect was naloxone-reversible and methylnaloxonium-irreversible proving that central opiate receptors must be solely responsible for mediating the induced analgesia (Fig. 10). It could be concluded, therefore, that the peptide CDS successfully delivered, retained, and released the peptide in the brain.

Modifications of the spacer (S) and lipophilic (L) components greatly influenced the efficacy of the CDS package proving that these components have important roles in determining molecular properties

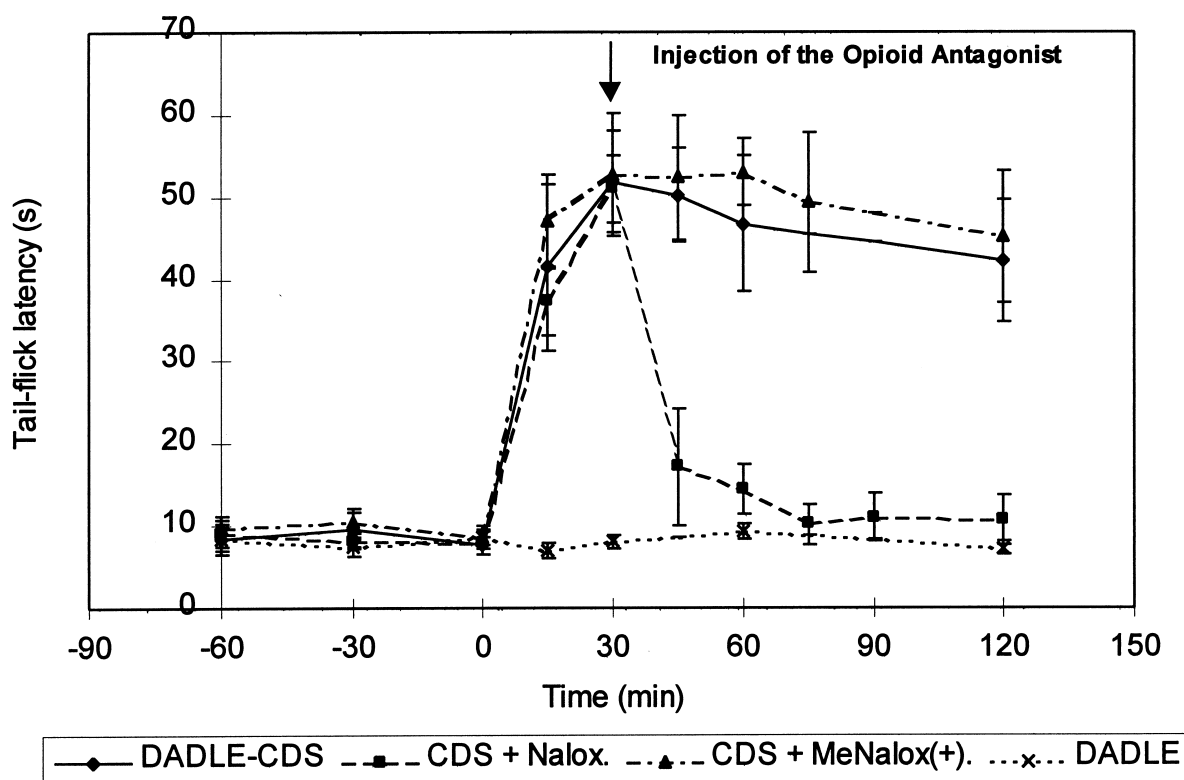


Fig. 10. Reversal of the analgesia produced in rats by 4.2 $\mu\text{mol/kg}$ DADLE-CDS administered i.v. by naloxone (2.7 $\mu\text{mol/kg}$, s.c.) but not by methylnaloxonium (2.1 $\mu\text{mol/kg}$, s.c.), a quaternary analog of naloxone that cannot readily cross the blood–brain barrier.

and timing of the metabolic sequence. The bulkier cholesteryl group used as L showed a better efficacy than the smaller 1-adamantaneethyl, but the most important factor for manipulating the rate of peptide release and the pharmacological activity turned out to be the spacer (S) function: proline as spacer produced more potent analgesia than alanine.

5.3. TRH analogue

A similar strategy was used to deliver a thyrotropin-releasing hormone (TRH) analogue to the central nervous system [132]. These analogues are potential agents for treating neurodegenerative disorders such as Alzheimer's disease. Because the peptide that has to be delivered has no free -NH_2 and -COOH termini, a precursor sequence (Gln–Leu–Pro–Gly) that will ultimately produce the final TRH analogue was packaged for delivery (Fig. 9). Therefore, two additional steps had to be included in the metabolic sequence: one where the C-terminal adjuster glycine is cleaved by peptidyl glycine α -amidating monooxygenase (PAM) to form the ending prolinamide, and one where the N-terminal pyroglutamyl is formed from glutamine by glutaminyl cyclase.

Selection of a suitable spacer proved also important for the efficacy of TRH-CDSs as measured by the decrease in the barbiturate-induced sleeping time in mice. After i.v. administration of equimolar doses of 30 $\mu\text{mol/kg}$, the TRH analogue itself produced only a marginal decrease, while the CDS with L-Ala spacer produced a $\sim 30\%$ reduction, and the one with a double alanine spacer produced a greater than 50% decrease [157].

5.4. Kyotorphin analogues

Finally, a kyotorphin analogue (Tyr–Lys) that has activity similar to kyotorphin itself was also successfully delivered [133,134]. Kyotorphin is an endogenous neuropeptide (Tyr–Arg) that exhibits analgesic action through the release of endogenous enkephalin, and its analgesic activity is about 4 times larger than that of Met-enkephalin [160]. As this peptide contains a free amine residue, and since preliminary studies indicated such ionizable functions might prevent successful delivery, this free amine function-

ality was additionally 'packaged' by attachment of a Boc (*tert*-butoxycarbonyl) group. It was assumed that this attachment is bioconvertible, and the 'lock-in' mechanism will allow time for its enzymatic removal.

Since during the synthesis of this CDS package, α - and ϵ -amine functions had to be differentially functionalized, combined liquid phase Fmoc and Boc chemistry was employed as summarized in Fig. 11. With this synthetic method at hand, a number of various spacers were explored for this delivery system (proline, proline–alanine, and proline–proline), and amongst them the double proline provided the best pharmacological effect. It was found that even with the single proline spacer the corresponding CDS showed good activity in the rat tail-flick latency test. Activity was already significant at a 3 $\mu\text{mol/kg}$ (1.0 mg/kg) dose and leveled off at about 22 $\mu\text{mol/kg}$. This represents an improvement of about two orders of magnitude compared to the ~ 200 mg/kg dose necessary to observe activity when the peptide itself is given intravenously. Several intermediates and building blocks were also studied, but only administration of the whole molecular package produced significant pharmaceutical response (Fig. 12) confirming that only the designed metabolic sequence as a whole is effective in delivering peptides across the BBB.

Another significant result achieved here was the design and first evaluation of a novel and conceptually different method that can provide brain-targeted activity for peptides containing amino acids with basic side chains (lysine, arginine). Within this approach, the targetor T moiety, which by conversion to its charged T^+ form is responsible for the 'lock-in' mechanism, is not attached to the peptide from 'outside', but it is integrated within novel 'redox' amino acids building blocks that replace the original basic amino acid of the active peptide. Consequently, isosteric/isoelectronic analogy between the original and the 'redox' side-chain and not cleavage of the redox targetor is expected to provide activity. Fig. 13 compares the structure of the 'classical' brain-targeted CDS with the structure of the novel brain-targeted redox analog (BTRA) used for the present kyotorphin analogue. As shown in Fig. 14, both the CDS and the BTRA produced significant and prolonged analgesic effect in rats, and

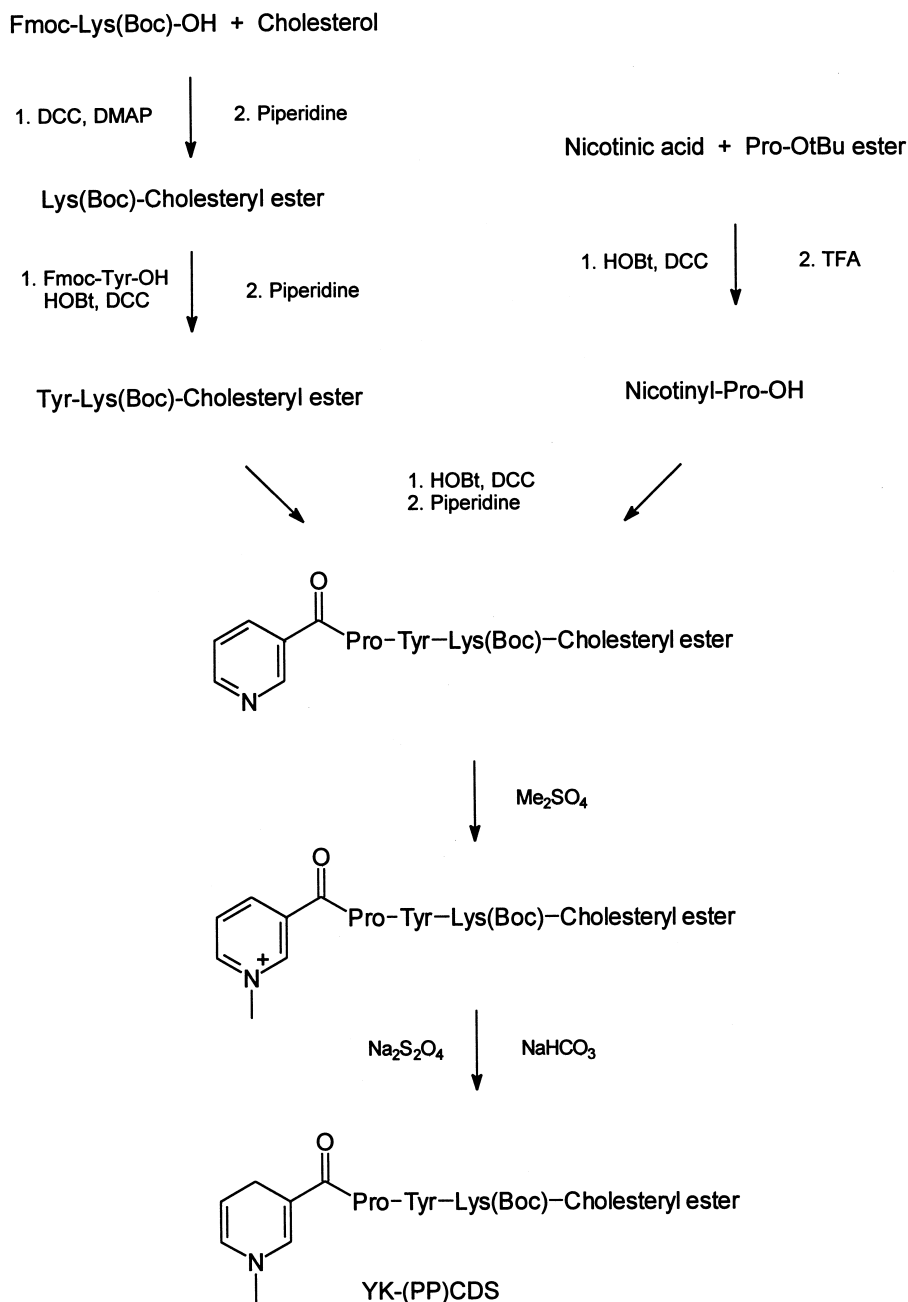


Fig. 11. The synthesis of the kyotorphin analogue (YK) CDS. As both α - and ϵ -amine functions had to be functionalized, combined liquid phase Fmoc and Boc chemistry was employed. Notation used: Boc, (*tert*-butoxycarbonyl); DCC, dicyclohexylcarbodiimide; DMAP, (dimethylamino)pyridine; Fmoc, fluorenylmethyloxycarbonyl; HOBt, 1-hydroxybenzotriazole; TFA, trifluoroacetic acid.

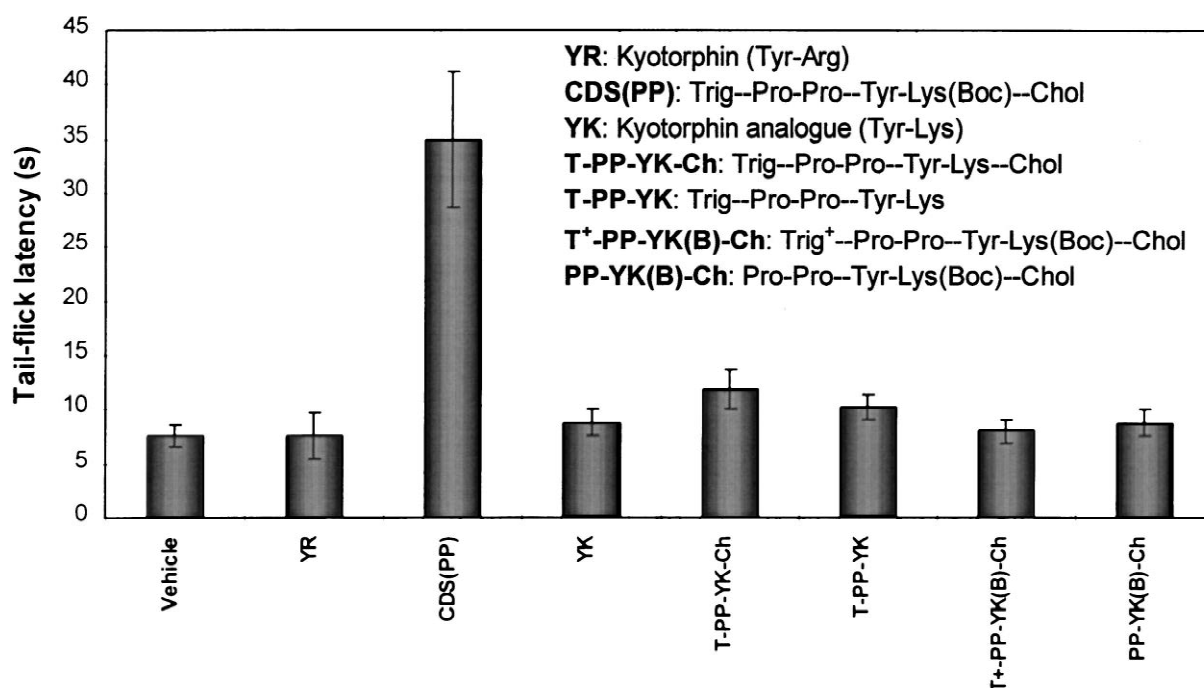


Fig. 12. Analgesia produced in rats by i.v. administration of 22.3 $\mu\text{mol/kg}$ kyotorphin (YR), its analogue (YK), the chemical delivery system with a double proline spacer (CDS(PP)), and several intermediates and building blocks. Only administration of the whole CDS package produced a pharmacologically significant response.

both could be reversed by naloxone demonstrating again that central opiate receptors must be responsible for mediating the induced analgesia. Such novel amino acids may provide both effective replacement and targeting for other lysine- or arginine-containing peptides.

In conclusion, molecular packaging of peptides is a rational drug design approach that achieved the first documented noninvasive brain delivery of these important biomolecules in pharmacologically significant amounts. By overcoming the obstacles imposed by the BBB, this approach may represent an important step toward future generations of high efficiency neuropharmaceuticals obtained from biologically active peptides.

6. Computer-aided design

Because of the importance of physicochemical (i.e., partition) properties during the sequential metabolism, knowing how these properties relate to

molecular structure is of great interest. However, because these latest CDSs are quite large molecules and often contain peptide constituents, they are difficult to handle with most available computer methods, including semiempirical quantum chemical methods.

However, our recently developed, molecular size-based method [137–139,161] allows integration of some quantitative physicochemical aspects into CDS designs. The model can estimate octanol/water partition properties for a wide variety of organic molecules including nonzwitterionic peptides and some large compounds where usual two-dimensional fragmental methods completely fail. It is of remarkable simplicity and uses as parameters only the computed van der Waals molecular volume (V) and a newly introduced parameter (N) that might describe changes in hydrogen bonding during the octanol–water transport.

After implementation of the corresponding computer program, its predictive power was tested on a set of 438 molecules with a variety of structures

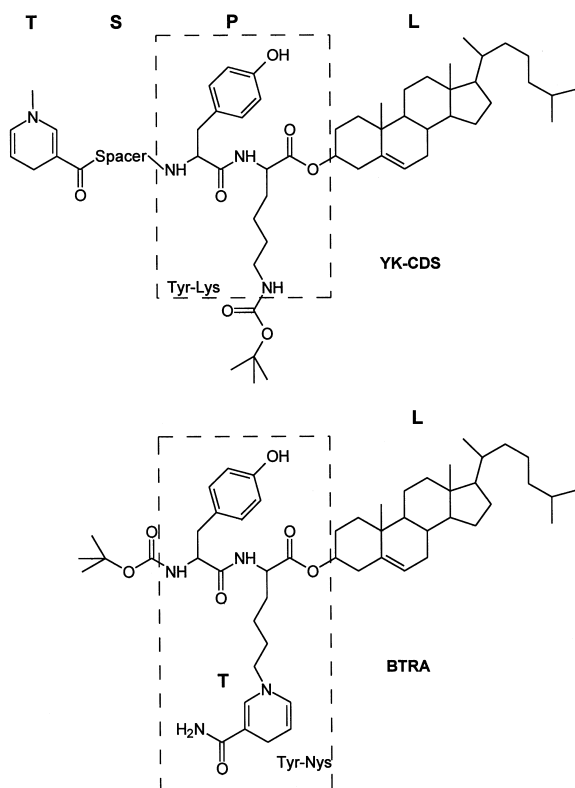


Fig. 13. The chemical delivery system (CDS) and the brain-targeted redox analog (BTRA) used for the delivery of the kyotorphin analogue YK (spacer: Pro, Pro–Pro, or Pro–Ala).

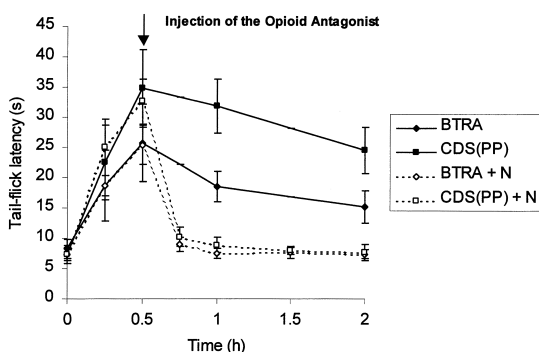


Fig. 14. Reversal of the analgesia produced by 22.3 $\mu\text{mol/kg}$ YK-CDS and BTRA administered i.v. by naloxone (6.1 $\mu\text{mol/kg}$, s.c.) 30 min later. Data represent mean $\pm$ SE of six rats for each group. The tail-flick latency periods for CDS and BTRA were not recorded at 0.75 and 1.5 h.

[137]. The correlation obtained between predicted and experimental values was quite impressive for such a simple model on such a wide range of compounds: $r=0.975$, $\sigma=0.365$, $F=8575.8$, slope=1.0023, intercept=0.0025. As a further and promising development, the same model could be extended to give good correlation for a set of 188 neutral peptides ($r=0.956$, $\sigma=0.455$, $F=1978.0$, slope=1.066, intercept=−0.166) [138]. In going to these structures, only two reasonable and probably hydrogen-bonding-related adjustments had to be introduced to the N -contribution of the hydrogen-bearing amide functions and the aliphatic hydroxyl-containing side-chains, respectively. Predictions were in realistic range even for larger peptides (cyclosporin, melanotan, octreotide) where overparametrized group contribution methods (CLOGP, ACD/LogP) proved quite unreliable. For example, cyclosporin A has $\log P_{\text{exp}}=2.96$, $\log P_{\text{calc}}=1.28$, CLOGP=14, ACD/LogP=11.3 and octreotide has $\log P_{\text{exp}}=1.45$, $\log P_{\text{calc}}=1.51$, ACD/LogP=−3.91. It also seemed possible to extend the present model to estimate apparent partition coefficients measured around neutral pH and physiological conditions for monoionic peptides, but since such partition data are buffer-dependent, the situation here is more complex [138].

The same model could also describe reasonably well the partition data of quaternary pyridinium-type compounds, which contain a nitrogen moiety similar to that used in the present targetor moiety, by using a value of $N=9$ for quaternary pyridinium nitrogens: $n=18$, $r=0.964$, $\sigma=0.424$, $F=207.7$ [140]. Compared to the neutral form ($N=2$ for pyridine), this represents a significant change of about five orders of magnitude ($7 \times 0.72 \approx 5$)—quite consistent with a successful ‘lock-in’ mechanism.

An example of such calculated properties for compounds formed during the sequential metabolism of a CDS is provided in Fig. 5 for lomustine-CDS. Similar data for E_2 -CDS were summarized earlier [63]. Predictions for large and special compounds always have to be treated with caution, but for one example of brain-targeting peptide CDS, experimental measurements confirmed our prediction. The distribution coefficient at pH 7.0 for the quaternary form of a TRH-CDS with a double proline spacer could be determined ($\log D_{\text{exp}}=0.25$), and was in

good agreement with the predicted value ($\log D_{\text{calc}} = 0.82$) [140]. These developments allow inclusion of quantitative estimates into future CDS designs.

7. Conclusions

In conclusion, the CDS approach can provide brain-targeted delivery and sustained release for a variety of neuropharmaceuticals. Since the approach represents a general and systematic method that also allows for considerable flexibility in the design, it can be applied for almost any lead compound. A number of concrete examples where successful delivery and sustained activity have been achieved were provided. They clearly prove that, with adequate design, the approach can provide substantially increased and prolonged brain exposure. In addition, molecular packaging, an extension of the CDS approach, achieved the first documented noninvasive brain delivery of neuropeptides in pharmacologically significant amounts.

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